

Heterogeneous Effects of Internal and External Built Environments on LST Under Different Weather Conditions: A case study of Beijing, China

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Abstract

Rapid urbanization and climate change have led to a sharp rise in temperatures in urban neighborhoods, where land surface temperature (LST) directly affects residents' thermal comfort and health. While existing studies have predominantly focused on the impacts of internal neighborhood features on thermal environments, the influences of external environments and weather conditions remain insufficiently examined. This study proposes a refined framework incorporating both the internal and external environments of neighborhoods. We retrieved summer LST data under sunny, heatwave, and cloudy conditions in Beijing. Using multi-source geographic data, we applied random forest (RF) and multiscale geographically weighted regression (MGWR) models to quantify the relative importance and spatial heterogeneity of internal and external built environment factors. Results show that (i) Incorporating external environmental factors of 90 m significantly increases the performance of models; (ii) External tree canopy coverage and building coverage are key factors correlating with neighborhood average LST; (iii) The spatial heterogeneity patterns of internal and external factors are different. These findings provide scientific insights for neighborhood renovation practices and have practical implications for advancing sustainable neighborhood development and improving human settlement environments.

Keywords:

Built environments, Land surface temperature, Neighborhoods, Random Forest, Spatial heterogeneity

1. Introduction

In 2024, the global average annual temperature surpassed the critical 1.5°C threshold above pre-industrial levels for the first time on a full-year basis, underscoring the intensifying climate crisis (World Meteorological Organization, 2025). This global warming, coupled with rapid urbanization, profoundly exacerbates the Urban Heat Island (UHI) effect in metropolises like Beijing, where summer temperatures in urban areas can be up to 4°C higher than in rural regions (Meng et al., 2018). The intensity of the UHI is increasing at a rate of 0.24°C per year (Ge et al., 2016), with extreme heat gradients becoming significantly steeper towards the urban core (Hou et al., 2025). As the frontline of climate impact, urban neighborhoods expose residents, particularly vulnerable populations like the elderly and children, to heightened heat-related health risks (Chen et al., 2024; Lao et al.,

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2025), increased energy burdens (Yang et al., 2020), and potential displacement (Lao et al., 2025). Therefore, conducting neighborhood-scale research on the thermal environment in metropolis is imperative for sustainable urban development.

Currently, extensive research has established that a neighborhood's internal built environment, including land use, 3D building morphology, and green space patterns, significantly influences its thermal environment (Chen et al., 2022; Sun et al., 2025). Liu et al. (2024a) studied the nonlinear relationship between built environmental factors and LST in different Local Climate Zones (LCZs). Cheng et al. (2025) compared the relative importance of different environmental factors on LST. Liu et al. (2026) found that the effect of the built environment on LST exhibits spatial heterogeneity within Haidian District, Beijing. However, these works have predominantly focused on internal features, largely overlooking the potential influence of the external built environment of the neighborhood. Establishing a thermal environment analysis framework that incorporates both internal and external environments can help us comprehensively understand the relationship between the neighborhood built environment and thermal environment. Current research has also noticed the effect of the external environment. Nevers et al. (2025) found the non-local cooling effect of trees surrounding neighborhoods. Lin et al. (2025) also found that external environmental factors are correlated with LST variation within different LCZs. However, most existing studies have regarded the buffer distances as a fixed (Lin et al., 2025) or experiential value and have only evaluated the contribution or influencing direction of the external environmental factors (Kang et al., 2026; Zawadzka et al., 2021). A deeper understanding is still needed regarding the distances of buffer zones, their relative importance, nonlinear effects, and the spatial heterogeneity of the external environment. Understanding the effect of the external environment comprehensively is crucial for a holistic understanding of the neighborhood thermal environment and for developing comprehensive cooling strategies. The common omission of these factors in neighborhood-scale models thus hinders accurate thermal environment assessment and attribution.

Existing studies have also revealed that the effects of thermal environment factors are not static. Instead, they are influenced by meteorological factors. Zhang et al. (2025b) pointed out that the cooling effect of green spaces is most pronounced on LST in summer. Wang et al. (2023b) found that the influence of urban characteristics on LST is related to solar radiation intensity. Fang et al. (2025) documented that trees and lawns exhibit distinct cooling patterns during heatwaves. Chen et al. (2025a) proposed optimizing green space layout to adapt to heatwave conditions. However, few existing studies have examined the impacts of urban characteristics on the thermal environment across different weather scenarios. Most studies have only focused on the role of key factors under a single weather condition (e.g., sunny days, heatwaves), and the research conducted under a single weather condition has limitations in explaining different weather scenarios (Song et al., 2025). Beijing's temperate continental monsoon climate, with its highly variable summer conditions that alternate among sunny, cloudy, and heatwave days (Zhang et al., 2025a), allows for a critical examination of how meteorological conditions modulate thermal environmental factors. Cross-scenario comparisons are necessary to robustly verify the stability of internal and external factors and to understand the neighborhood thermal environment across different weather conditions. Therefore, we need to understand the role of environmental factors in the thermal environment of neighborhoods under different weather conditions.

Furthermore, the relationship between a neighborhood's thermal environment and the year of construction warrants further investigation. Residential neighborhoods built at different times in Beijing exhibit significant differences in tree cover, building density, and building height (Zheng et al., 2017). However, few studies have examined the relationship between neighborhood thermal environments and construction years. Beijing, as a northern Chinese metropolis, offers an ideal laboratory to address these gaps (Liu et al., 2024a). Its unprecedented urbanization has created a mosaic of neighborhood forms across different construction years, from old, low-rise, high-density quarters to modern, high-rise, open neighborhoods (Yu et al., 2026). Clarifying the role of the construction year is also essential for understanding the thermal disparities between old and new neighborhoods and for devising targeted renovation strategies (Su et al., 2022).

To address these research gaps, this study investigates the impact of internal and external built environments on neighborhood LST in Beijing under varying weather conditions, with a specific focus on the distances of buffer zone. First, using multi-source geographic data, we compare the performances of models with different buffer zones under various scenarios. Then, we employ a combination of random forest (RF) (Salman et al., 2024), Shap-

ley Additive Explanations (SHAP) (Zhang et al., 2025c; Zhu et al., 2025), and Partial Dependence Plots (PDP) (Cheng et al., 2025) to quantify relative importance and reveal the nonlinear relationship of key environmental factors. Finally, we apply Multiscale Geographically Weighted Regression (MGWR) (Huang et al., 2022; Zhong et al., 2024) to explore the spatial heterogeneity of these factors under sunny, cloudy, and heatwave scenarios. The overall framework of this study is shown in Fig. 1. Specifically, this study aims to address the following research questions: (i) Does incorporating buffer zone variables improve the performance of predicting LST in neighborhoods? What buffer distance yields optimal model performance? (ii) How do internal and external built environment characteristics contribute to the neighborhood thermal environment under different weather scenarios, and what are the differences in their relative contributions and nonlinear relationships? (iii) Do the effects of internal and external environmental characteristics exhibit spatial heterogeneity? Do the spatial heterogeneity of internal and external environmental factors perform the same? Through the above analyzes, this study seeks to unravel the relationships between internal and external built environments and the thermal environment of neighborhoods in Beijing. The findings are expected to provide a scientific basis and replicable research paradigm for formulating neighborhood cooling strategies and optimizing urban livable environments.

The remainder of this paper is organized as follows. Section 2 introduces the research materials and methods, including study area delimitation, multi-source geographic data collection and pre-processing, variable screening, and the application of RF, SHAP, PDP, and MGWR models. Section 3 presents the research results, such as the optimal external buffer scale selection under different weather conditions, as well as the relative importance and spatial heterogeneity of internal and external built environment factors on LST. Section 4 discusses the selection of the buffer zone, spatial spillover effects, and spatial heterogeneity of key factors, as well as their practical implications for Beijing's neighborhood planning, and also points out the study limitations and future research directions. Section 5 summarizes the core conclusions of the study on the impacts of internal and external built environments on neighborhood LST and expounds on the implications of the findings for urban heat island mitigation and sustainable neighborhood development.

2. Materials and Methods

2.1. Overall workflow of this study

The overall workflow of this study is illustrated in Fig.1, which consists of three main phases: acquiring data sources, selecting study variables, and comprehensive analysis.

The first and second steps can be collectively referred to as data preparation. We constructed a geographic information dataset of neighborhoods and established a comprehensive variable framework. Detailed procedures can be found in Section 2.3. The third step is data analysis. First, we built Random Forest(RF) models and compared the influence of external environmental characteristics at different buffer distances on model performance, thereby selecting the optimal buffer distance to characterize the external environment of neighborhoods. Second, we constructed full models incorporating both internal and external environmental factors and used Shapley additive explanations to evaluate the relative importance of variables across the six dates, identifying the most influential factors. Subsequently, we conducted an in-depth analysis of these key factors. We visualized the marginal effects of the important factors using Shapley additive explanations and applied multiscale geographically weighted regression (MGWR) to examine their spatial heterogeneity and spatial distribution patterns. This approach provides urban managers and planners with a detailed strategy for identifying the roles of internal and external neighborhood environments and optimizing the neighborhood thermal environment.

2.2. Study area

Beijing is located in the North China Plain, with a geographic extent of 39°28' - 41°25' N and 115°25' - 117°30' E, covering a total area of approximately 16,800 km² (Chen et al., 2023). As the national capital, political center, and international hub of China, Beijing ranks among the world's most populous metropolises. Over the past half-century, it has undergone unprecedented urbanization. It's built-up area expanded from less than 100 km² to over 1,400 km², and the population surged from 7 million to more than 21 million by 2020 (Wang et al., 2023a). Beijing has a temperate continental monsoon climate, characterized by hot and rainy summers, resulting

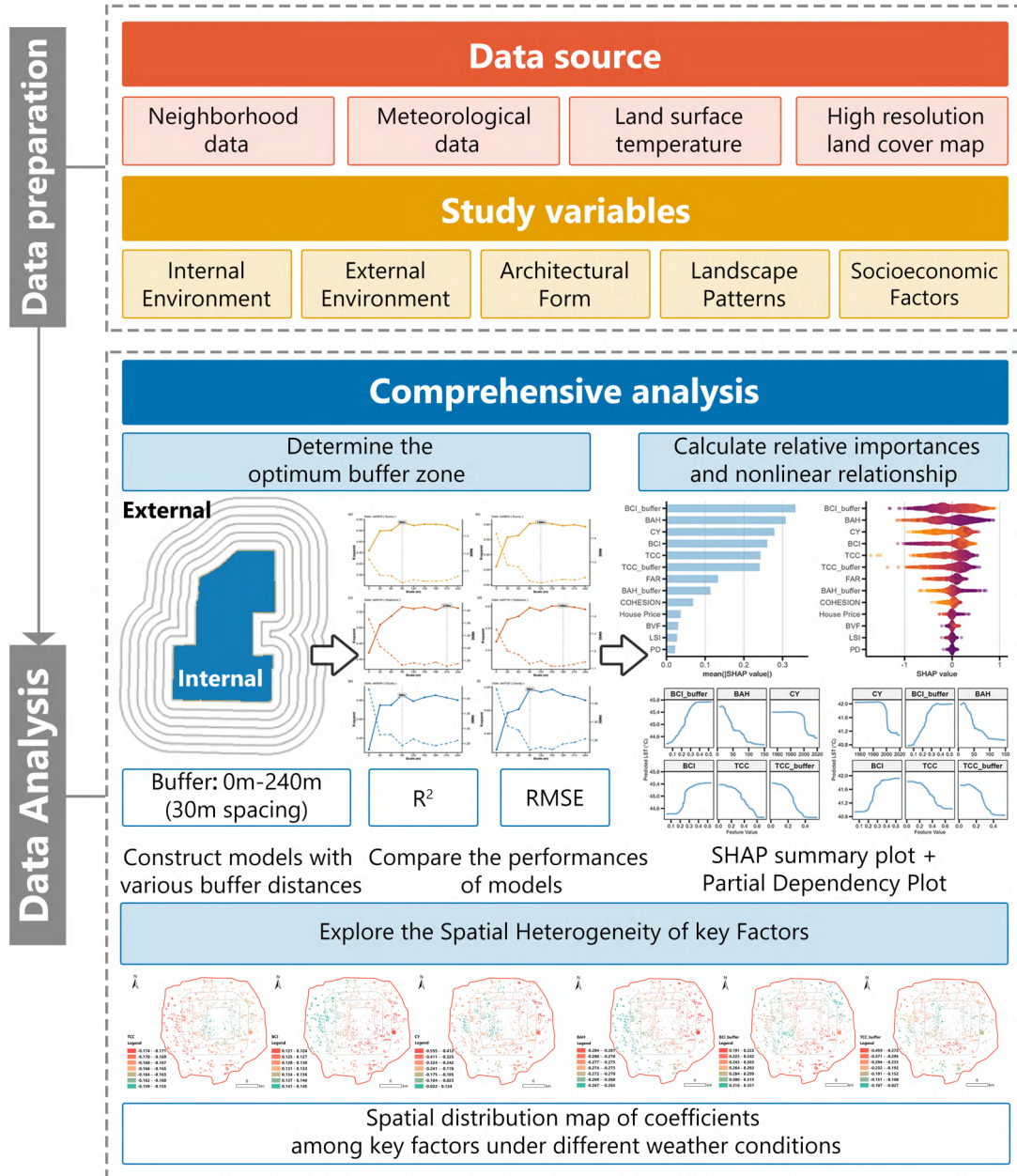


Figure 1: Overall workflow from data preparation to data analysis.

in a pronounced urban heat island (UHI) effect (Fang et al., 2024). As the fundamental spatial unit of human settlement, urban neighborhoods are directly exposed to UHI impacts (Jiang et al., 2024a). In this study, the study area was confined within Beijing's Fifth Ring Road, which is the core area of Beijing. This study covers 1,345 neighborhoods (Fig.2).

2.3. Data Sources

2.3.1. Neighborhood data sources

Neighborhood data were obtained from two sources, i.e., the Baidu Maps Open Platform(<https://lbs.baidu.com/>), which provided boundary vectors, and Anjuke(<https://www.anjuke.com>), a leading Chinese real estate platform that supplied attribute records. This website serves as a reliable source of data for research across China(He et al., 2025; Wu et al., 2018). First, we obtained Points of Interest (POIs) of 'residential neighborhood(zhuzhai xiaoqu in Chinese)' via the Baidu Maps JavaScript API and retrieved the unique identifier (UID) for each neighborhood. Subsequently, we extracted the neighborhood boundary coordinates from the Baidu Maps

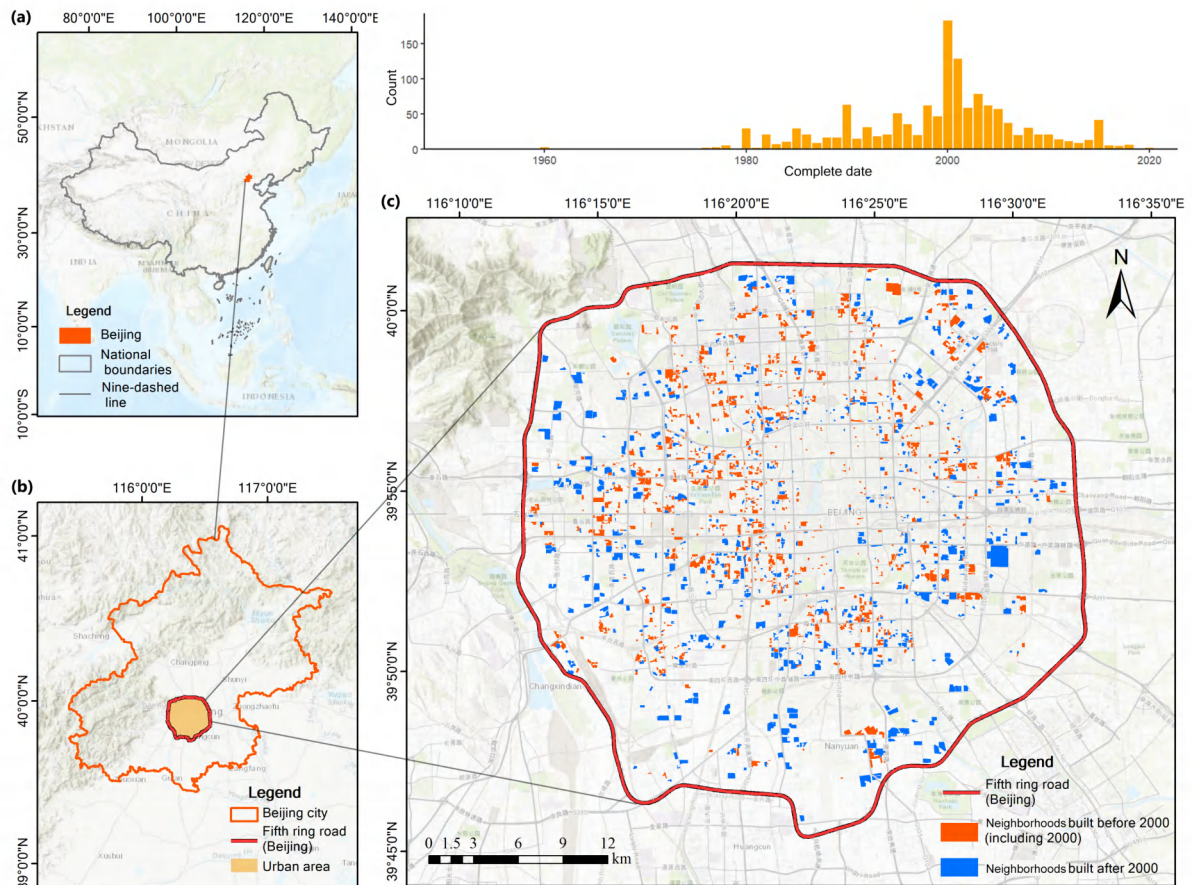


Figure 2: Beijing's fifth ring road area and selected neighborhoods.(a) National boundary of the People's Republic of China (b) The boundary of Beijing. (c) 1,345 neighborhoods selected for the study

back-end interface using UID and converted the Baidu metric coordinates into standard WGS_1984_UTM_Zone_50N longitude and latitude through projection transformation. After cleaning and verifying the topology, we converted these geographic coordinates into standardized polygon vectors and saved them in a shape-file format that contains essential information such as neighborhood names, spatial boundaries, and geographic locations.

Second, these neighborhood AOI vectors were matched with the housing price and construction year data from Anjuke using a series of matching rules to ensure the consistency of multi-source data. We retrieved data on construction year and housing price from Anjuke to characterize the socioeconomic and developmental profiles of each neighborhood. To ensure data quality and spatial consistency, we applied a set of strict screening criteria: (i) Only neighborhoods with valid names that were successfully matched to the Anjuke database and with clearly documented construction years were retained; (ii) Only neighborhoods completed by 2020 were included, ensuring temporal consistency with the acquisition year of remote sensing datasets; (iii) Neighborhoods smaller than 0.3 ha were excluded to eliminate fragmented patches; (iv) Neighborhoods with an average building height exceeding 150 m were removed to comply with building codes and reflect actual urban morphology; (v) Neighborhoods located within 300 m of the 'Fifth Ring Road' boundary were excluded to prevent buffer overflow beyond the study area.

After geographic correction and filtering, this study retained a total of 1,345 valid neighborhoods (Fig. 2). Furthermore, in accordance with the Guiding Opinions of the General Office of the State Council on Comprehensively Promoting the Renovation of Old Urban Residential neighborhoods (General Office of the State Council of the People's Republic of China, 2020), we defined neighborhoods constructed in or before 2000 as older neighborhoods, while those completed after 2000 were classified as newer neighborhoods. Among the final sample, 725 were older neighborhoods, predominantly concentrated in the central and northwestern areas within the Fifth Ring Road, and 620 were newer neighborhoods, primarily distributed in peripheral zones.

2.3.2. Meteorological data

Meteorological data were collected from the official historical weather website(<https://lishi.tianqi.com/>) of China and the NASA POWER database (<https://power.larc.nasa.gov/>) to define three distinct weather conditions for the summer seasons spanning 2020 to 2023, including sunny days, heatwaves, and cloudy days. Sunny days were defined to include both completely clear skies and transitional conditions between sunny and cloudy. This broader definition was adopted due to the rarity of persistently clear skies in Beijing’s summer and the similarity of their surface warming patterns under direct solar radiation. Heatwaves were identified according to the official standards set by the National Meteorological Center of China and the Meteorological Disaster Warning Signals and Prevention Guidelines (China Meteorological Administration, 2007), which specify days with a daily maximum temperature of at least 35 °C that persist for three consecutive days or longer. Cloudy days, on the other hand, encompassed fully overcast conditions as well as transitions between cloudy and rainy weather. We first conducted a preliminary screening using weather data from official historical weather databases. Then, we used the ‘nasapower’ package in R version 4.5.3 to derive the data of meteorological parameters for the region centered at 39.90°N, 116.40°E to detect weather conditions in the core of Beijing.

2.3.3. LST data of neighborhoods

LST data were retrieved from Landsat Collection 2 Level-2SP products, which provide a Surface Temperature (ST) product. The ST data are stored in the band ST_B10. To properly interpret Landsat Collection 2 Surface Temperature (ST) data, we must apply a scaling factor to convert digital numbers (DN) to temperature values in degrees Celsius. Following the method provided on the USGS website (<https://www.usgs.gov/landsat-missions/landsat-collection-2-surface-temperature>), the equation is used:

$$LST = (DN * 0.00341802) + 149.00 - 273.15, \quad (1)$$

where the *DN* represents digital numbers of band ST_B10 in the Landsat Collection 2 Level-2 product.

In this study, we selected the high-quality LST image in Google Earth Engine(<https://earthengine.google.com/>). By screening meteorological conditions and taking into account the availability of LST imagery, six images were finally selected to represent sunny, heatwave, and cloudy weather conditions, respectively. Detailed meteorological parameters for the acquisition dates of these images are provided in Table S1 of the supplementary materials. As shown in Fig.S1, we selected six scenes to represent each weather condition: August 3, 2020, and June 3, 2021, for sunny days; June 19, 2021, and July 19, 2023, for heatwaves; and June 9, 2023, and July 24, 2022, for cloudy days. These dates correspond to different weather conditions. On sunny days and heatwaves, the LST imagery of the Fifth Ring Road area was not obscured by cloud cover. For cloudy scenes, we masked cloud and shadow contamination using the QA_pixel band provided with the Landsat datasets and subsequently excluded neighborhoods affected by cloud cover. This filtering resulted in 1,295 valid neighborhoods for the June 9, 2023, scene and 1,266 for the July 24, 2022, scene, representing a slight reduction from the 1,345 neighborhoods available under sunny and heatwave conditions. Finally, we calculated the mean LST within each neighborhood boundary and reported the values in degrees Celsius (Fig. 3(b)).

2.3.4. Built environment factors

First, to ensure that the dataset can accurately reflect the detailed land use of neighborhoods in Beijing, we created a high-resolution land use map applicable to the area within the Fifth Ring Road. The land cover map applied in this analysis is presented in Fig. 3(a). This map was developed by integrating several datasets, i.e., urban tree cover from Meta-WRI tree canopy data (Tolan et al., 2024), the built-up dataset from Baidu Maps (<https://map.baidu.com/>), and categories of artificial surfaces, grass and fields, and barren land from the ESRI 10-meter global land use data (<https://livingatlas.arcgis.com/landcover/>). To assess the accuracy of the fused land cover map, we constructed a confusion matrix using 300 manually labeled sample points. These points were collected from 0.6 m resolution satellite imagery acquired in June 2020 via the Bigemap Pro platform (<http://www.bigemap.com/>), with no fewer than 50 points per land cover class. The integrated product achieved an overall accuracy of 89.2%, confirming its suitability for community-scale research. A table of category-specific accuracy is presented in Table.S2. In addition, we also obtained the building height data of the study area from the

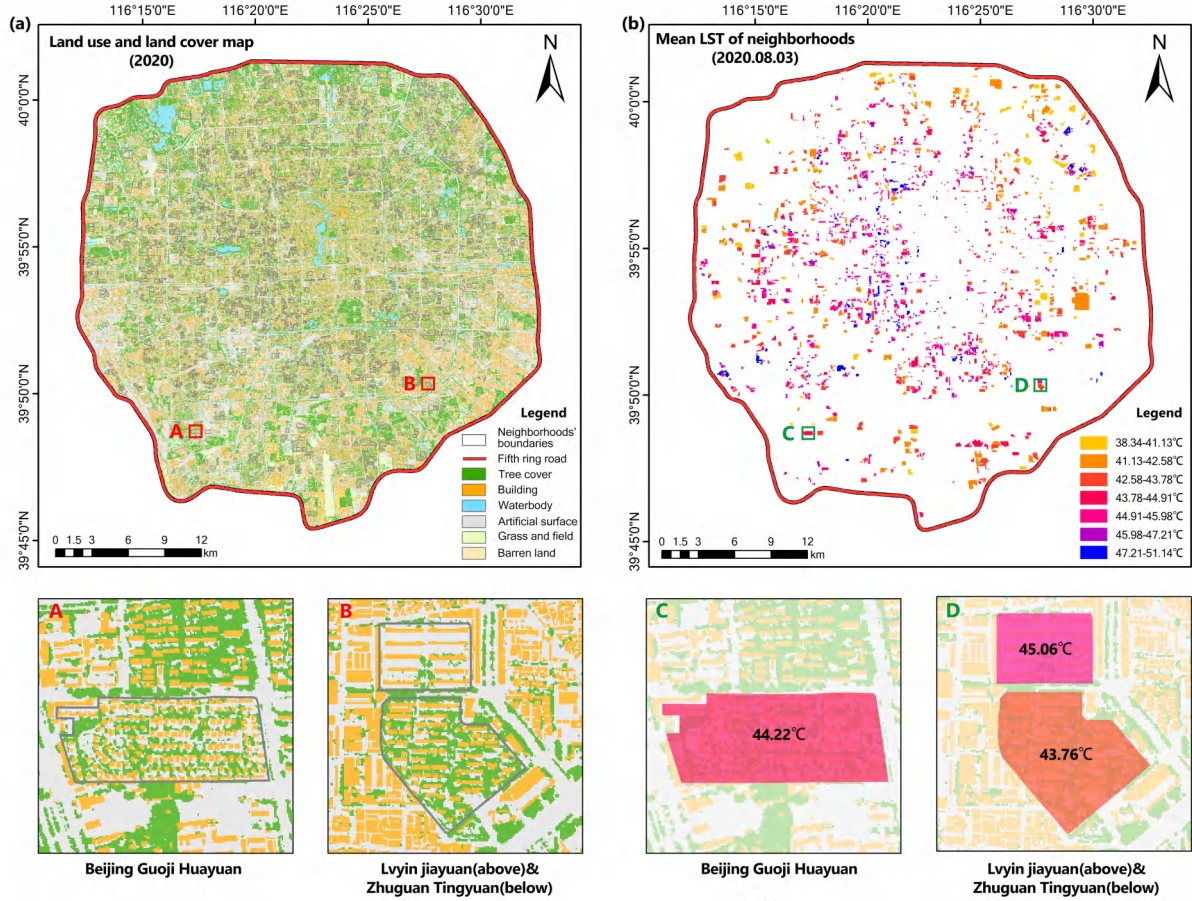


Figure 3: Land cover map of the fifth ring road area and average LST of each neighborhood. (a) The land use and land cover map of neighborhoods (b) Mean LST of each neighborhood on 3 August 2020.

Baidu Maps Open Platform (<https://lbs.baidu.com/>), which is suitable for calculating the 3D characteristics of neighborhoods.

Then, in this study, we identified a comprehensive set of variables, which we categorized into five main groups, i.e., internal built environment, external built environment, architectural form, landscape patterns, and socioeconomic factors (Table.1). For the internal built environment, we quantified three indicators: Tree Canopy Cover (TCC), Building Coverage Index (BCI), and Building Average Height (BAH). These represent, respectively, the proportion of tree cover, the building coverage ratio, and the average building height within each neighborhood boundary. Correspondingly, for the external built environment, we characterized three analogous indicators, TCC_buffer, BCI_buffer, and BAH_buffer, which refer to the tree canopy cover, building coverage, and average building height within the external buffer zone of each neighborhood. To describe architectural form, we employed the Floor Area Ratio (FAR) and the Building Sky View Factor (BVF). FAR reflects the overall development intensity of a neighborhood, while BVF quantifies the sky openness of buildings. We calculated BVF using building height data and a Digital Elevation Model (DEM) in SAGA GIS. For landscape patterns, we selected three metrics related to the tree canopy: Patch Cohesion Index (COHESION), Landscape Shape Index (LSI), and Patch Density (PD). These metrics have been widely validated and are known to be closely associated with the urban thermal environment (Sun et al., 2025; Liu et al., 2026). We calculated landscape patterns in Fragstats 4.2. We represented socioeconomic factors using two variables, including the neighborhood's construction year and its housing price. The construction year refers to the year the estate was finished, whilst the property prices are sourced from the Anjuke website, which provides the average asking price for properties sold in the estate in December 2020. These serve as proxies for the neighborhood's developmental stage and socioeconomic status.

Finally, we established a general linear model using the 13 independent variables and the LST under six dates and assessed the presence of severe multicollinearity among variables using the Variance Inflation Factor (VIF).

As shown in Tables S3–S7 in the supplementary file, all VIF values were less than 5, indicating no severe multicollinearity among the variables, which allows us to proceed with subsequent analyzes using random forest models.

Table 1: Indicators and calculations

Category	Indicator	Meaning	Formula / Source
Internal Environment	TCC	Measurement of tree canopy coverage in neighborhoods.	$TCC = \frac{S_{tree}}{S_{total}}$
	BCI	Measurement of building coverage index in neighborhoods.	$BCI = \frac{S_{building}}{S_{total}}$
	BAH	Measurement of building average height in neighborhoods.	$BAH = \frac{\sum_{i=1}^n H_i}{n}$
External Environment	TCC_buffer	Measurement of tree canopy coverage in the buffer zone.	$TCC_buffer = \frac{S_{TCC_buffer}}{S_{buffer}}$
	BCI_buffer	Measurement of building coverage index in the buffer zone.	$BCI_buffer = \frac{S_{BCI_buffer}}{S_{buffer}}$
	BAH_buffer	Measurement of building average height in the buffer zone.	$BAH_buffer = \frac{\sum_{i=1}^{n_b} H_{b_i}}{n_b}$
Architectural Form	FAR	Measuring the intensity of development in neighborhoods.	$FAR = \frac{\sum_{i=1}^n (S_i \times F_i)}{S_{total}}$
	BVF	Measuring the mean value of sky view factor of buildings in neighborhoods.	$BVF = \frac{\sum_{i=1}^n SVF_i}{n_p}$
Landscape Patterns	COHESION	Patch Cohesion Index of tree canopy in neighborhoods.	$COHESION = \left(1 - \frac{\sum p_i}{\sum p_i \sqrt{a_i}} \cdot \frac{1}{\sqrt{S_{tree}}}\right) \times 100$
	LSI	Landscape Shape Index of tree canopy in neighborhoods.	$LSI = \frac{0.25 \times E}{\sqrt{S_{tree}}}$
	PD	Patch Density of tree canopy in neighborhoods.	$PD = \frac{N}{\sqrt{S_{tree}}}$
Socio-economic Factors	CY	Construction year of neighborhoods.	"Anjuke" website
	House price	Price per square metre of house in the neighbourhood (¥/m²).	(https://beijing.anjuke.com/)

Note: S_{total} : total size of neighborhoods; S_{buffer} : area of the buffer zone external neighborhoods; H_i : height of single building in neighborhoods; n : the number of buildings in neighborhoods; H_{b_i} : height of single building in the buffer zone; n_b : the number of buildings in the buffer zone; S_i : area of each building in neighborhoods; F_i : number of floors of each building in neighborhoods; SVF_i : Sky View Factor's value for a single pixel; n_p : pixel count; E : total length (m) of edge in tree canopy between patch types; N : Total number of tree canopy patches in the landscape; p_i : perimeter of tree canopy patch i ; a_i : area of tree canopy patch i .

2.4. Methods

This study proposes an integrated framework combining Random Forest (RF) and Multiscale Geographically Weighted Regression (MGWR) to explore the impacts of internal and external factors on the neighborhood thermal environment. The methodology consists of three main steps. First, to select the optimal buffer distance, we iteratively incorporated external environmental features at different scales into RF models with fixed parameters and compared model performance. Second, we identified key variables with high relative importance using global SHAP values and further visualized their nonlinear relationships via Partial Dependence Plots (PDP). Third, we applied MGWR to model the spatial heterogeneity of these key variables. This systematic approach enables us to identify optimum buffer distances, key factors of neighborhoods' environments, and understand their effects on LST in neighborhoods.

2.4.1. Selecting the optimum buffer distances through Random Forest Modeling

In this study, the RF algorithm was employed to determine the optimal buffer scale and quantify the relative impacts of internal and external built environments on neighborhood LST. As an ensemble machine learning technique, RF is particularly effective for handling complex, nonlinear relationships and small-sample datasets (Liu et al., 2013). It constructs numerous decision trees during training and outputs the final prediction through majority voting, thereby ensuring high prediction stability and generalization capability (Kim and Kim, 2022). To identify the optimal buffer distance, we also trained a series of fixed-parameter RF models (with $n_{tree} = 1000$ and min node

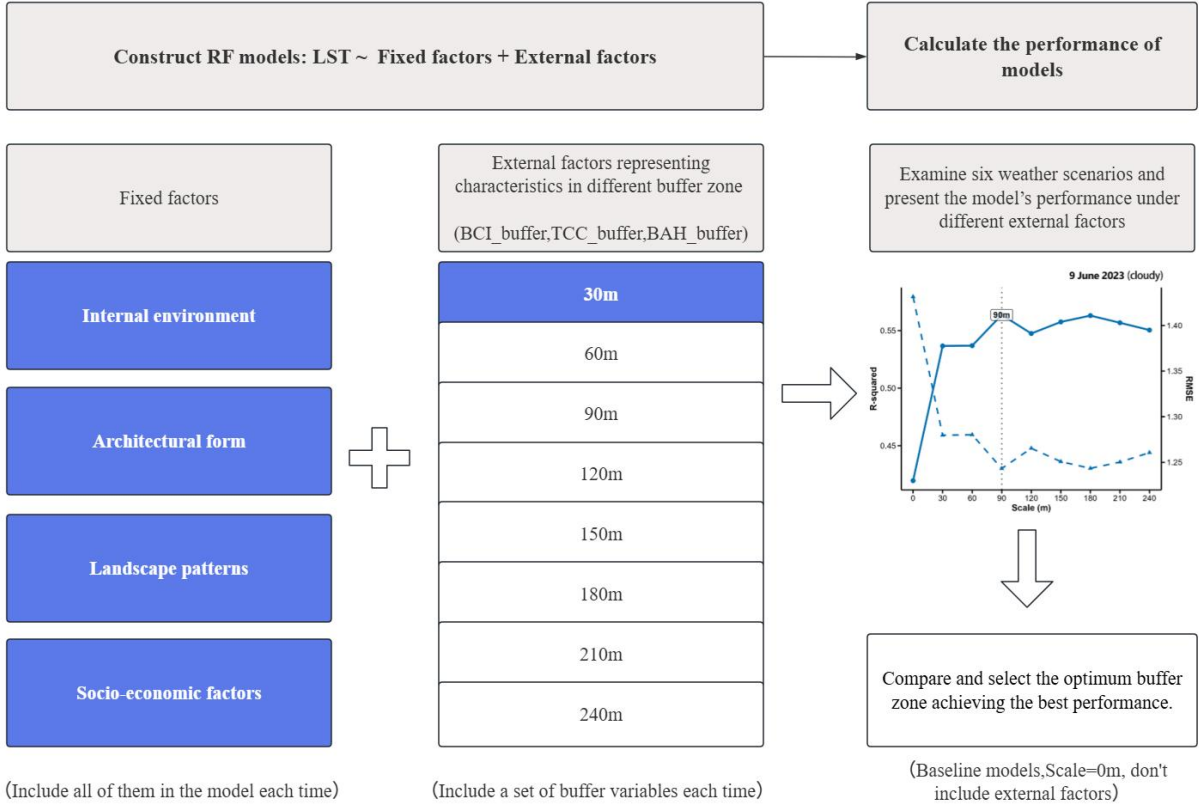


Figure 4: Flowchart of selecting the optimum external factors

size = 5) using buffer sizes ranging from 0 m to 240 m at 30 m intervals. The performance of each model, measured by R^2 and Root Mean Square Error (RMSE), was evaluated using five-fold cross-validation. The external characteristics with optimum performance will be applied in the further study, and we will model the LST separately for each date. This process is shown in Fig. 4. Bayesian optimization was applied to fine-tune the hyperparameters, including ntree (ranging from 100 to 2000), mtry (1 to 13), and min.node.size (1 to 9), with RMSE serving as the loss function. Five-fold cross-validation was again utilized to ensure model robustness. Finally, we conducted Shapley Additive Explanations (SHAP) analysis on the optimized RF model to interpret the model predictions.

2.4.2. Interpreting contributions and nonlinear relationships of factors based on SHAP and PDP

In this study, we applied Shapley Additive Explanations (SHAP), a game theory-based approach, to interpret the predictions of the optimized RF model (Shapley, 1953). SHAP values provide a unified framework for quantifying the contribution of each feature to individual predictions, enabling the assessment of both relative feature importance and the direction (positive or negative) of their influence. Through this analysis, we were able to summarize both global and local interpretability. We derived global feature importance from the mean absolute SHAP values, while visualizing the direction of each feature using SHAP summary plots. The Shapley value for a feature is formally defined as the following formula:

$$\text{Shapley}(X_j) = \sum_{S \subseteq N \setminus \{j\}} \frac{k!(p-k-1)!}{p!} (f(S \cup \{j\}) - f(S)), \quad (2)$$

where p is the total number of features, $N \setminus \{j\}$ is a set of all possible combinations of features excluding X_j , S is a feature set in $N \setminus \{j\}$, $f(S)$ is the model prediction with features in S , and $f(S \cup \{j\})$ is the model prediction with features in S plus feature X_j .

To further elucidate complex nonlinear relationships between the top-ranked predictive features and LST, we generated partial dependence plots (PDP) (Friedman, 2001). PDP illustrates the marginal effect of one or two features on the model's predicted outcome (Wang et al., 2025), while averaging out the effects of all other features. In this study, we applied PDPs to visualize the six most influential variables and identify their potential marginal

impacts on LST across different weather conditions (sunny, heatwave, cloudy).

2.4.3. Exploring the spatial heterogeneity of factors based on MGWR

In this study, we employed multiscale geographically weighted regression (MGWR) to explore the spatial heterogeneity effects of the key factors. Unlike traditional regression models, MGWR allows each explanatory variable to have its own unique optimal bandwidth, thereby enabling the detection of spatial instability across multiple spatial scales. We implemented the MGWR algorithm in ArcGIS Pro. First, we conducted global spatial autocorrelation analysis for LST data acquired on the six dates. Significant spatial clustering of LST justifies the use of geographically weighted regression models. Given the scattered spatial distribution of neighborhoods across the study area, we set the neighborhood type to 'number of neighboring features' and adopted the 'golden-section search method' for bandwidth optimization to guarantee model rationality.

3. Results

3.1. Descriptive analysis of neighborhoods' characteristics

In this study, we present descriptive statistics for LST and built environment characteristics across neighborhoods, as summarized in Table 2. We observed that the mean LST within neighborhoods varied substantially across different weather conditions, ranging from 31.099°C to 52.320°C. The standard deviations (SD) of neighborhood mean LST remained relatively consistent, fluctuating between 1.523°C and 1.946°C, which suggests stable distribution patterns under various summer weather scenarios. Regarding the built environment characteristics, we found that the mean values of Tree Canopy Coverage (TCC) and Building Coverage Index (BCI) were 0.312 and 0.294, respectively. The mean value of Building Average Height (BAH) is 45.5 m, with a considerable SD of 23.839 m, reflecting a wide morphological diversity spanning low-rise to high-rise urban forms. The mean Floor Area Ratio (FAR) and Building View Factor (BVF) were 5.123 and 0.493, respectively. Additionally, the median construction year (CY) and house price were 2000 and 85,994 RMB/m², respectively. In terms of landscape patterns, we identified that the mean value of the Landscape Shape Index (LSI) is 1.052, with a small SD, suggesting relatively regular patch shapes within neighborhoods. Conversely, the Patch Cohesion Index (COHESION) and Patch Density (PD) demonstrated substantial variability, with SDs of 12.522 and 238.472, respectively.

Table 2: Mean LST in neighborhoods and neighborhoods' built environment.

Variable	Min-Max	Mean	Median	SD
<i>LST Indicators (°C)</i>				
Sunny (3 June 2021)	31.099–40.089	35.960	35.981	1.523
Sunny (3 August 2020)	38.629–51.138	44.574	44.689	1.946
Heatwave (19 June 2021)	41.715–52.320	47.088	47.131	1.801
Heatwave (19 July 2023)	40.744–50.756	45.958	46.060	1.775
Cloudy (9 June 2023)	37.280–50.626	45.072	45.172	1.884
Cloudy (24 July 2022)	36.269–46.988	41.654	41.734	1.828
<i>Neighborhoods' built environment</i>				
TCC	0.009–0.715	0.312	0.312	0.125
BCI	0.049–0.544	0.294	0.289	0.073
BAH(m)	6.000–150.000	45.500	38.333	23.839
FAR	0.132–15.181	5.123	4.449	2.459
BVF	0.256–0.995	0.493	0.482	0.094
CY(A.D.)	1952–2020	1998.658	2000	9.456
House Price(¥/m ²)	4170–283908	86913.675	85994.000	32706.138
LSI	1.000–1.333	1.052	1.050	0.028
COHESION	0.000–99.621	66.612	68.534	12.522
PD	4444.444–8888.889	4663.181	4632.835	238.472

In this study, we employed Spearman's rank correlation analysis to examine the relationships among neighborhood characteristics, with the results presented in Fig.S2. We observed a strong positive correlation between

COHESION and TCC ($r = 0.81$), suggesting that neighborhoods with higher tree canopy coverage tend to exhibit better canopy connectivity. Additionally, we identified a significant positive correlation between LSI and PD ($r = 0.66$), indicating that complex patch shapes are frequently associated with more fragmented landscape patterns. Regarding architectural attributes, we also found a strong positive correlation between FAR and BAH ($r = 0.79$), implying that higher floor area ratios are associated with taller buildings. Conversely, BAH exhibited negative correlations with both BCI ($r = -0.41$) and BVF ($r = -0.53$), suggesting that increased building height is correlated with decreased building coverage and building view factor. In addition, we detected a significant negative correlation between BCI and TCC ($r = -0.40$), reflecting a competitive trade-off between building footprint and tree canopy coverage within limited neighborhood space.

3.2. Optimal distances of buffer zone for RF models

To investigate the influence of external built environment features on neighborhood thermal conditions, we developed RF models incorporating variables of external built environmental characteristics across multiple buffer ranges. As illustrated in Fig. 5, we observed similar performance trends across models, although the optimal buffer distances varied according to weather conditions. Regarding the similar trend, our results demonstrate that models including external environmental variables, namely TCC_buffer, BCI_buffer, and BAH_buffer, exhibited substantially improved performance compared to models excluding these features. Specifically, the buffer distance of 90 m is a turning point characterized by a notable rise in R^2 and a corresponding decrease in RMSE. Beyond this threshold of 90 m, we observed that fluctuations in both R^2 and RMSE remained relatively modest. Regarding the optimal buffer distance, we found that for sunny and cloudy summer conditions, the appropriate buffer range was 90~120 m. Interestingly, during heatwave conditions, we detected the optimal buffer ranging from 180~210 m. Based on these findings, we established final RF models and conducted hyperparameter optimization according to the specific optimal buffer ranges identified for each weather scenario. The optimized model was used to assess the relative importance of variables and non-linear relationships.

3.3. Contributions and nonlinear relationships of factors

3.3.1. Relative importance and contribution of characteristics

We use the SHAP method to evaluate the relative importance of variables. Fig. 6 (a)~(f) illustrates the relative importance and direction of influence of various driving factors under different weather scenarios. Across all six scenarios, BCI_buffer, BCI, CY, BAH, TCC, and TCC_buffer consistently ranked as the top six influencing factors. Among these, BCI_buffer emerged as the most important variable in most scenarios. The relative importance of TCC_buffer was comparable to that of internal tree canopy coverage (TCC), highlighting the substantial and stable influence of the external built environment on the LST of neighborhoods. Although the construction year (CY) is not a direct physical characteristic of the environment, it exhibited high contribution. Internal factors such as building average height (BAH), TCC, and building coverage index (BCI) also played critical roles in contributing to the LST. In contrast, external building height (BAH_buffer) and internal landscape pattern indices characterizing tree canopy morphology (COHESION, LSI, PD) showed relatively weaker importance. Regarding the direction of influence, both internal and external building coverage (BCI and BCI_buffer) exhibited a positive correlation with LST, whereas internal and external tree canopy coverage (TCC and TCC_buffer) and building heights (BAH and BAH_buffer) demonstrated negative correlations. Among socioeconomic variables, CY showed a negative relationship, while house prices displayed mixed or negative correlations. With respect to architectural form, floor area ratio (FAR) negatively correlated with LST, whereas BVF had a positive relationship. For internal tree canopy configuration, COHESION showed a clear negative relationship, LSI exhibited a weak negative relationship, and PD yielded mixed influences across scenarios.

Fig. 6 (g) presents the contribution rates of different factor categories across the six scenarios. In terms of total contribution, internal built environment factors (BCI, TCC, BAH) accounted for 37.7%, slightly higher than the 33.0% contributed by external environmental factors (BCI_buffer, TCC_buffer, BAH_buffer). Together, these two categories explained 70.7% of the total importance. The remainder was attributable to socio-economic factors (15.5%), architectural form characteristics (7.9%), and neighborhood landscape pattern indices (6.0%).

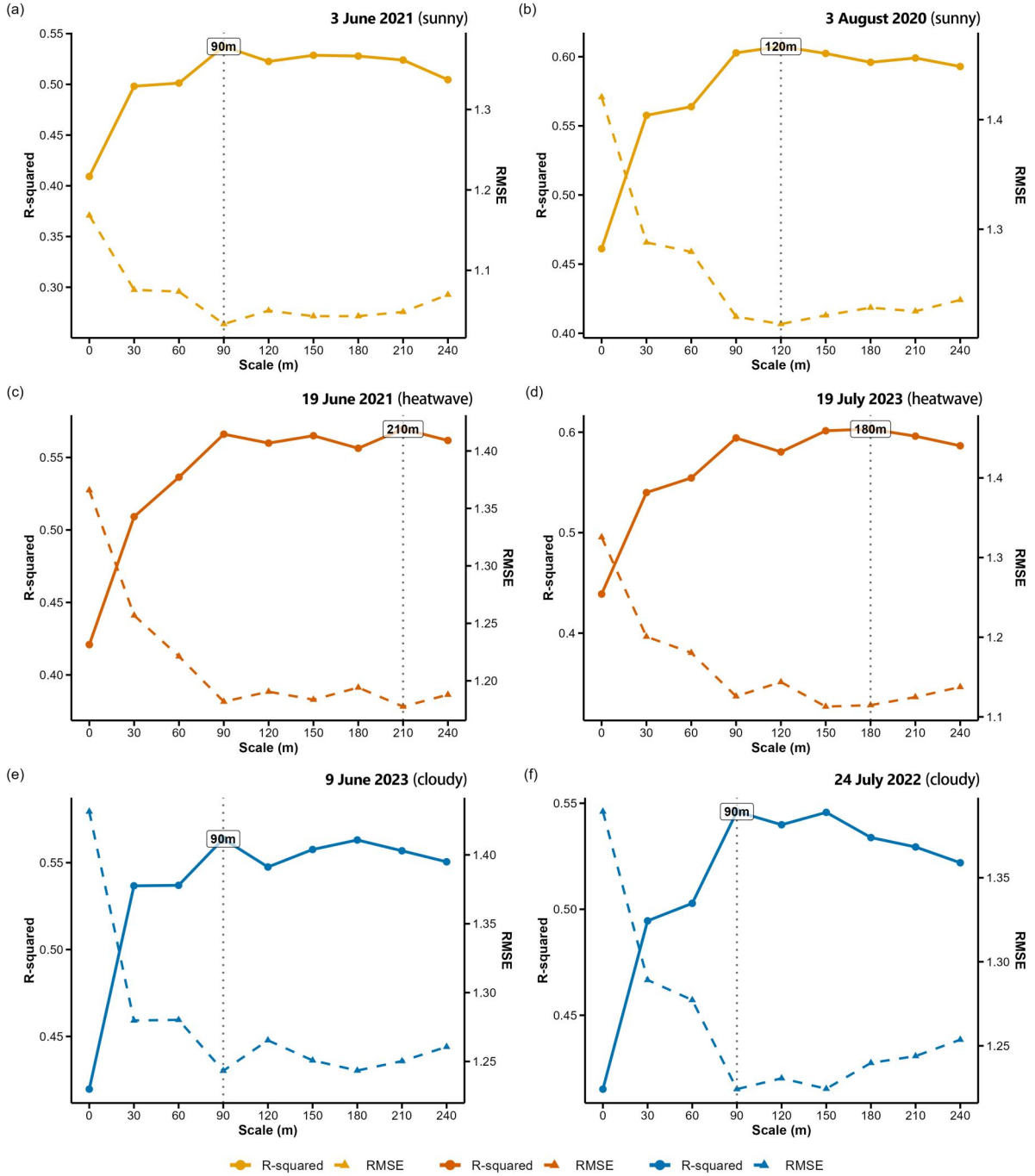


Figure 5: Performance comparison of the random forest (RF) models under diverse weather conditions. The numbers in the boxes indicate the optimal buffer distance. Orange curves represent sunny days, red curves represent heatwaves, and blue curves represent cloudy days. (a) 3 June 2021; (b) 3 August 2020; (c) 19 June 2021; (d) 19 July 2023; (e) 9 June 2023; (f) 24 July 2022.

In summary, BCI_buffer, BCI, CY, BAH, TCC, and TCC_buffer were identified as the key characteristics contributing to summer neighborhood LST. Both internal and external environmental attributes were found to be critical determinants, with external building coverage exerting a strong positive relationship and external tree canopy providing a negative relationship. The CY also emerged as an important predictor of LST. Understanding the contributions and directions of these factors enhances the understanding of how internal and external neighborhood characteristics jointly shape the thermal environment.

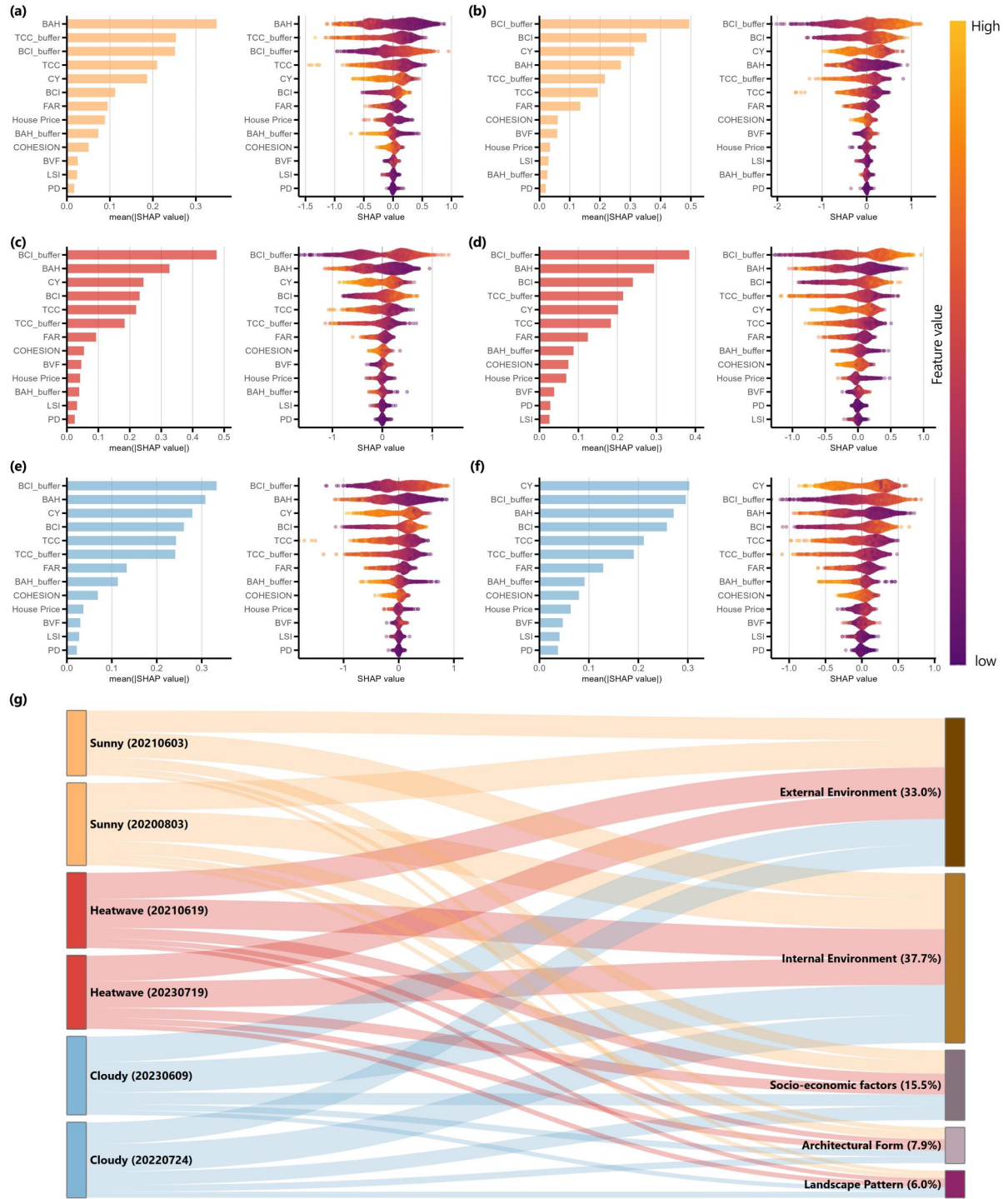


Figure 6: Relative contribution and influencing direction of variables. Orange bars represent sunny days, red bars represent heatwaves, and blue bars represent cloudy days. (a) SHAP summary plot for LST prediction on 3 June 2021 (sunny); (b) SHAP summary plot on 3 August 2020 (sunny); (c) SHAP summary plot on 19 June 2021 (heatwave); (d) SHAP summary plot on 19 July 2023 (heatwave); (e) SHAP summary plot on 9 June 2023 (cloudy); (f) SHAP summary plot on 24 July 2022 (cloudy); (g) Contribution of different factor categories to LST under diverse weather conditions.

3.3.2. Nonlinear relationship of the important factors on LST

In this study, we used the SHAP method to identify the six most important factors of neighborhood LST across different scenarios. The result is shown in Fig. 7 (a)–(f). The results indicate that BCI_buffer and BCI generally show positive relationships with LST, whereas CY, BAH, TCC, and TCC_buffer exhibit predominantly negative relationships, with slight variations observed across different weather conditions.

Both internal and external building coverage demonstrates pronounced nonlinear warming relationships with

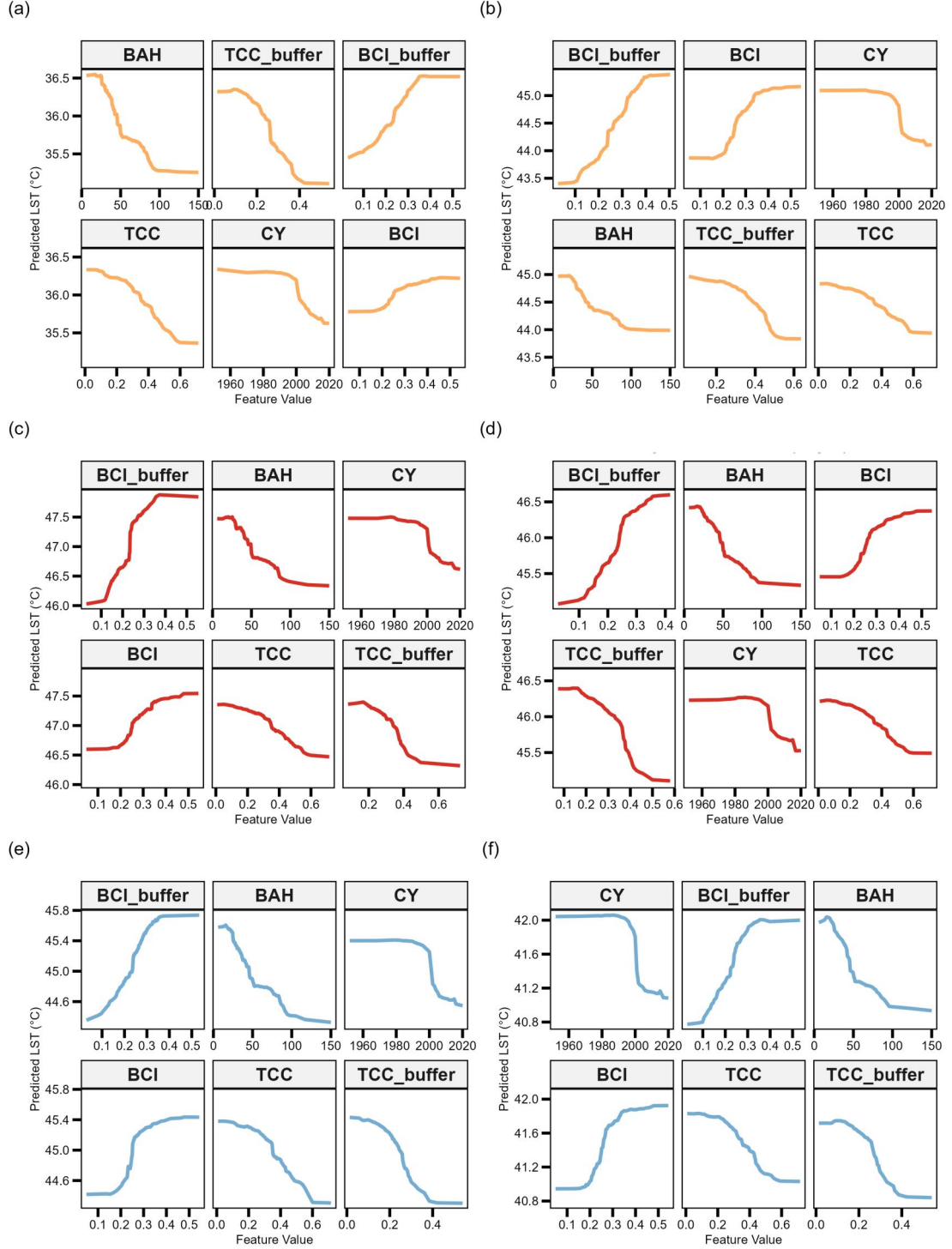


Figure 7: Nonlinear relationships of the top 6 variables with LST under diverse weather conditions. Orange curves represent sunny days, red curves represent heatwaves, and blue curves represent cloudy days. (a) 3 June 2021 ; (b) 3 August 2020 ; (c) 19 June 2021 ; (d) 19 July 2023 ; (e) 9 June 2023 ; (f) 24 July 2022.

1 LST, characterized by distinct response patterns. Specifically, BCI_buffer exhibits a marginal effect on LST : once
2 BCI_buffer exceeds a certain threshold, LST tends to stabilize. In contrast, the relationship between BCI and LST
3 approximates a logistic growth curve. LST increases rapidly only after BCI reaches a threshold, after which the
4 rate of increase gradually diminishes. For example, as illustrated in Fig. 7 (b), changes in LST become negligible
5 when BCI_buffer ≥ 0.4 . Similarly, when BCI ≥ 0.2 , LST rises sharply, but the rate of increase slows considerably
6 after BCI ≥ 0.4 .

In terms of cooling effects, TCC, TCC_buffer, and BAH exhibit nonlinear relationships with LST, wherein LST decreases as these variables increase, eventually stabilizing beyond certain thresholds. As shown in Fig. 7 (d), BAH is negatively correlated with LST, and LST remains relatively stable when $BAH \geq 100$ m. Likewise, Fig. 7 (c) demonstrates that LST declines with increasing TCC_buffer or TCC, but stabilizes when $TCC_buffer \geq 0.4$ or $TCC \geq 0.6$. A notable threshold is observed between CY and LST. As depicted in Fig. 7 (a)–(f), LST decreases sharply when $CY \geq 2000$, indicating a marked difference in thermal conditions between newer and older neighborhoods. Specifically, older neighborhoods tend to sustain higher LST levels during the summer, underscoring the role of construction year as a critical contributor to the thermal environment in neighborhoods.

3.4. Spatial heterogeneity analysis based on MGWR

A summary of the MGWR model results is presented in Table.S8, revealing distinct spatial heterogeneity in the effects of the six key factors on neighborhood LST. The analysis demonstrates that internal built environment factors (TCC, BCI, BAH) exhibit relatively weak spatial heterogeneity, with bandwidths large enough to cover most of the study area. For instance, the bandwidths for TCC and BAH reached 1266 on July 24, 2022, while those for TCC, BCI, and BAH reached 1345 on August 3, 2020, effectively covering the entire sample size. This suggests that the influence of internal built environment characteristics remains relatively consistent across the study area. In contrast, external environmental factors, TCC_buffer and BCI_buffer, demonstrated stronger spatial heterogeneity. On July 19, 2023, for example, the bandwidth of TCC_buffer was 340, substantially smaller than the 623 recorded for TCC, indicating that external environmental features exhibit more pronounced spatial variation compared to their internal counterparts. Additionally, construction year (CY) also displayed strong spatial heterogeneity, with bandwidths less than or equal to 100 in some scenarios. It displays stronger spatial variability.

The coefficients of the key factors are illustrated in Fig. 8 and Fig. 9. TCC exhibits a consistently negative relationship across the study area, as indicated by low coefficients and high bandwidth. A low coefficient area is located between the Fourth and Fifth Ring Roads, particularly in the northwestern sector. Conversely, weaker negative relationships, characterized by higher coefficients, are predominantly located in the southwestern area within the Third Ring Road, a region corresponding to the core urban area. The BCI demonstrates a stable positive relationship throughout the study area, with slightly higher coefficients observed in the northwest compared to the southeast. Building average height (BAH) displays a negative relationship globally. The coefficient is higher in the western area within the Third Ring Road and lower in the area situated between the Fourth and Fifth Ring Roads.

External building coverage (BCI_buffer) exhibits moderate spatial heterogeneity and primarily functions as a warming factor. Under non-heatwave conditions, a stronger positive relationship is observed in the northeast and southwest between the Second and Fifth Ring Roads. During heatwaves, the spatial heterogeneity of BCI_buffer intensifies, with a positive relationship in the eastern and southwestern areas between the Second and Third Ring Roads. External tree canopy coverage (TCC_buffer) also shows moderate spatial heterogeneity but mainly functions as a cooling factor. It exhibits lower coefficients in the southwestern region between the Second and Fourth Ring Roads. Higher coefficients are found in the northeastern region within the Fifth Ring Road and the southwestern region between the Fourth and Fifth Ring Roads.

CY reveals pronounced spatial heterogeneity. The coefficient in the eastern region within the Fifth Ring Road is lower than that in the western region. Specifically, neighborhoods in the eastern area between the Fourth and Fifth Ring Roads exhibit lower coefficients, whereas the western area within the Third Ring Road shows higher coefficients or even positive associations with LST.

4. Discussion

4.1. Buffer distance under different weather scenarios

This study found that incorporating landscape features within a 90 m radius of the neighborhoods into the model significantly improved the performance of the RF model. This conclusion is consistent with some previous studies. For instance, the cooling effect of trees has been documented to extend approximately 100 m (Nevers et al., 2025). Similarly, research indicates that the thermal influence of any land-use type exhibits a gradient effect, with its impact diminishing beyond approximately 100 m (RoohaniQadikolaie et al., 2025). Zawadzka et al. (2021) also

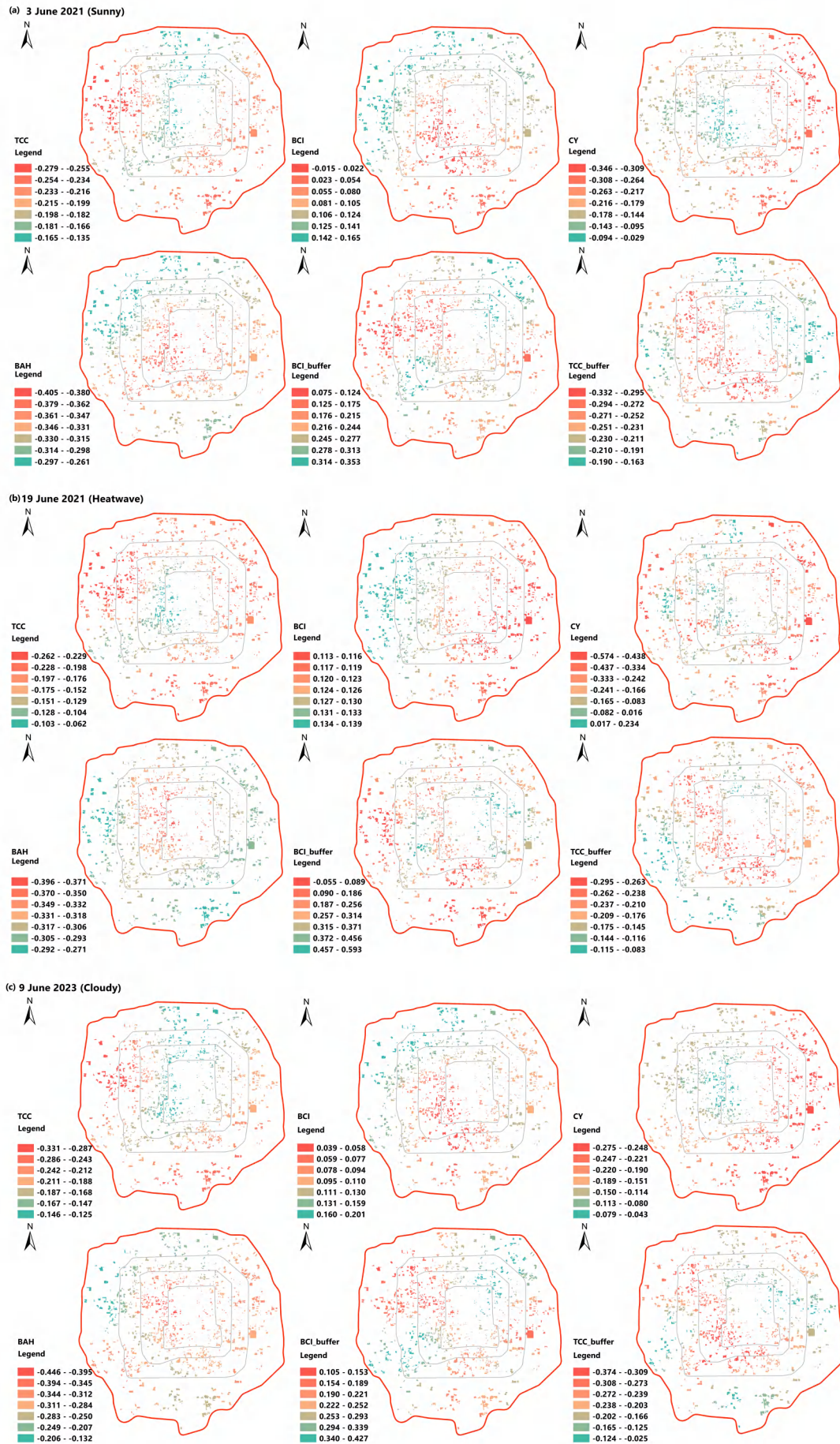


Figure 8: The spatial distribution of estimated coefficients for key factors in 3 scenarios.(a) 3 June 2021;(b) 19 June 2021;(c) 9 June 2023.

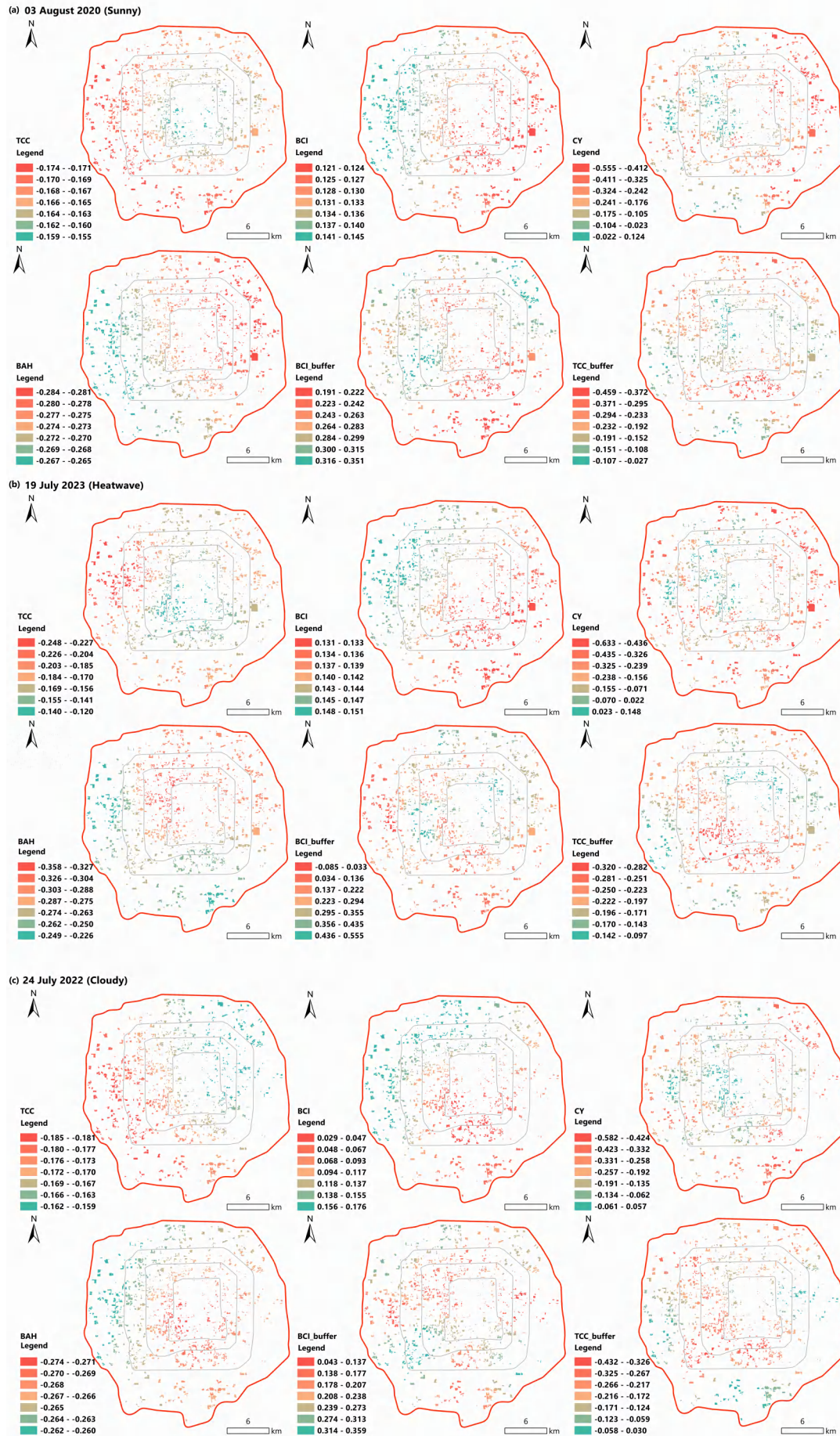


Figure 9: The spatial distribution of estimated coefficients for key factors in 3 scenarios.(a) 3 August 2020;(b) 19 July 2023;(c) 24 July 2022.

found that incorporating a buffer zone of 100 m enables RF models to achieve optimum performance in explaining LST in neighborhoods. Our findings corroborate these studies, demonstrating that a 90 m buffer sufficiently captures the external landscape characteristics that contribute to the thermal conditions within a neighborhood during the summer daytime.

Furthermore, our analysis revealed that the optimal buffer distance is not static but varies with weather conditions. As detailed in Fig.5, the optimal range expanded to 180–210 m during heatwaves, which is nearly double that of sunny or cloudy days (90–120 m). Previous research has demonstrated that landscapes play different roles during heatwaves. During heatwaves, built-up areas absorb and store more heat, subsequently releasing it as long-wave radiation and anthropogenic heat, which can propagate thermal influences over longer distances (Roshan et al., 2024). Concurrently, the cooling extent of green spaces has also been shown to increase under heatwave conditions (Wang et al., 2026). Moreover, heatwaves present unique weather conditions that may also affect the impact of the buffer zone. Heatwaves are often accompanied by meteorological conditions such as calm winds and temperature inversions (Li and Bou-Zeid, 2013), which restrict vertical heat diffusion and promote horizontal heat transport. This result demonstrates that during heatwaves, landscape features within a wider buffer zone are required to achieve optimal model performance. This finding highlights the importance of the landscape surrounding neighborhoods. We need to incorporate the wider buffer zone surrounding neighborhoods into our planning to address the threat of extreme weather.

4.2. Spatial spillover effects of external built environment

This study provides evidence of the significant spatial spillover effects of external built environments on neighborhoods' thermal environment. Our SHAP analysis consistently identified external building coverage (BCI_buffer) and external tree canopy coverage (TCC_buffer) among the top six most influential factors across all weather scenarios. In most cases, BCI_buffer contributed most substantially to LST variation, confirming the strong and dominant warming spillover effect of external buildings. This finding aligns with Chen et al. (2025b), who noted that the thermal effect of adjacent three-dimensional landscapes can be up to three times stronger than that of internal features. They attribute this to the building surface area-to-volume ratio (BSA/BV), which governs the absorption and emission of thermal radiation; a higher BSA/BV facilitates greater radiative heat release (Chen et al., 2025b). Similarly, we observed a consistent cooling spillover effect from external tree canopy coverage (TCC_buffer). This result corroborates the work of Lin et al. (2025), who demonstrated that vegetation external LCZs exert a stable cooling influence on internal LST throughout the diurnal cycle. Nevers et al. (2025) also reveal the non-local cooling effect of trees, whose cooling effect can extend up to approximately 100 m. This conclusion offers a new perspective on the management of the thermal environment in neighborhoods. We can not only regulate the internal thermal environment by managing the built environment within neighborhoods, but also improve it by adjusting the external environment—including increasing tree canopy cover and reducing building coverage.

4.3. Spatial heterogeneity of key factors

This study revealed marked differences in the degree of spatial heterogeneity among key factors of neighborhood LST. Our MGWR bandwidth analysis demonstrated that internal built environment variables, namely TCC, BCI, and BAH, exhibited relatively weak spatial heterogeneity compared to external built environment variables (TCC_buffer, BCI_buffer). In contrast, construction years (CY) exhibit stronger spatial heterogeneity across scenarios, reflecting discontinuous spatial patterns of neighborhood development and intensive human modifications. This finding is consistent with the conclusions of existing research. Gao et al. (2020) reveals that building coverage, building average height, and the Normalized Difference Vegetation Index (NDVI) exhibit global features in urban functional zones. Huang et al. (2025) finds that the cooling effect of horizontal greening (GVI) shows spatial stability. Notably, the bandwidth of BCI_buffer shrinks sharply during heatwaves, indicating intensified spatial heterogeneity. This may be attributed to the nature of heatwave conditions. Under heatwave conditions, stagnant airflow and temperature inversions (Li and Bou-Zeid, 2013) trap heat locally, enhancing spatially variable long-wave radiation emissions from external buildings and strengthening localized heating effects.

Moreover, external and internal factors present different spatial patterns in the study area. For example, tree canopy coverage in neighborhoods (TCC) shows higher coefficients in the urban core area (within the Third Ring

Road) compared to the marginal area (between the Fourth and Fifth Ring Roads). Current research also confirms that high-density development and intensive human activities weaken the cooling efficiency of urban greening (Guo et al., 2026). However, tree canopy coverage surrounding neighborhoods (TCC_buffer) exhibits lower coefficients in the southwest of the urban core area compared to the marginal area. This may be partly due to the presence of large parks (Tiantan park) in these areas, and partly because these areas are located on Beijing's main ventilation corridors in summer. Existing studies have shown that strengthening the connection between blue and green spaces and integrating the layout of green infrastructure with ventilation corridors can enhance the cooling effect of green infrastructure (Shen et al., 2026; Luo et al., 2024). This finding provides new insights into cooling neighborhoods in urban core areas. First, the cooling effect of internal tree canopy coverage is weakened in the urban core area; simply increasing tree canopy coverage within neighborhoods may hardly achieve satisfactory cooling. Second, external tree canopy coverage achieves higher cooling efficiency in the southwestern core areas and can be considered a more effective measure for cooling neighborhoods in urban core regions.

4.4. Policy implications for neighborhood renewal

First, improving the neighborhood thermal environment requires attention not only to the internal environment but also to the surrounding external environment. Under extreme summer weather (heatwaves), special attention should be given to the 200 m perimeter around the neighborhood, which should be planned and designed as a microclimate buffer zone to reduce the negative impacts of extreme weather and enhance residents' well-being.

Then, a striking threshold effect is detected for construction year (CY), with 2000 serving as a critical turning point: predicted land surface temperature (LST) drops sharply when $CY \geq 2000$ (7). The findings of this study offer actionable guidance for China's ongoing old neighborhood renovation campaign. Given that neighborhoods constructed prior to 2000 are explicitly targeted for urban renewal (General Office of the State Council of the People's Republic of China, 2020), special attention should be directed toward improving their thermal environments through targeted retrofitting strategies.

Moreover, many inner-city older neighborhoods face strict redevelopment constraints, and internal canopy expansion alone is often insufficient to achieve substantial cooling due to spatial limitations (Guo et al., 2024). The results underscore the value of an integrated internal-external planning approach. Enhancing external green spaces can effectively reduce internal LST, with particularly high cooling efficiency observed for external tree canopy coverage (TCC_buffer) in inner southwestern Beijing (Fig.8 and Fig.9). Beyond internal retrofitting measures, such as increasing greenery (Zeng et al., 2026), installing shade structures (Wu et al., 2025), applying high-albedo materials (Liu et al., 2024b), and implementing vertical greening (Jiang et al., 2024b), cooling benefits can also be attained through strategic upgrades to the external environment. These include creating pocket parks (Ma et al., 2023), expanding street greenery (Lee et al., 2025), demolishing dilapidated adjacent buildings (Karimi et al., 2025) to re-purpose land for green space, and increasing roadside tree canopy cover.

Such non-local effects provide a feasible and effective strategy for mitigating heat in densely built, inner-city older neighborhoods. To maximize cooling efficiency, external landscape interventions should be coordinated with local meteorological conditions and urban ventilation corridors, thereby enhancing airflow and optimizing the spatial distribution of cooling benefits.

4.5. Limitations and recommendations

This study provides a comprehensive analysis of internal and external built-environment effects on summer LST using multi-source high-resolution data. However, several limitations should be acknowledged.

First, accurately mapping vegetation and impervious surfaces beneath tree canopies via remote sensing remains challenging, potentially underestimating low vegetation and artificial surfaces. Future research could integrate UAV (Zhou, 2025) and LiDAR data (Münzinger et al., 2022) to refine neighborhood-scale landscape characterization.

Second, water bodies and certain landscape metrics were excluded due to data scarcity and multicollinearity. The limited blue infrastructure in Beijing results in an imbalanced sample, limiting generalization. Future studies in water-rich cities could better quantify the cooling effects of blue-green systems, and additional metrics could be included if multicollinearity is addressed by advanced variable selection.

Third, additional control variables should be considered to enhance comparability across models, including annual variations, antecedent humidity, cloud dynamics, wind regimes, and other background environmental factors.

Fourth, this study focuses only on summertime LST. Future work should capture seasonal and diurnal variations and incorporate air temperature and humidity measurements to better assess human thermal comfort.

Fifth, we observe that the spatial heterogeneity of BCI_buffer increases sharply under heatwave conditions, warranting further investigation. Future research could use computational fluid dynamics (CFD) models to simulate heat release from external buildings during heatwaves, providing a more scientific explanation.

5. Conclusion

This study proposes a novel framework for investigating the relationship between external and internal environmental factors and neighborhood thermal conditions. Focusing on 1,345 neighborhoods within Beijing's Fifth Ring Road, it integrates multi-source, high-resolution geographic data with LST observations under varying weather conditions. By employing RF algorithms and MGWR, the research systematically examines the optimum buffer distances, relative importance, nonlinear relationships, and spatial heterogeneity of internal and external built environments on neighborhood LST across different weather scenarios. The key innovative findings are threefold.

First, under sunny, heatwave, and cloudy daytime conditions in summer, this study found that the performance of models incorporating external buffer zone environmental characteristics was significantly improved. A buffer distance of 90 m was identified as a turning point; beyond 90 m, model performance remained relatively stable. Under heatwave conditions, the optimum buffer distances range from 180 m to 210 m, which is higher than those on sunny and cloudy days (90 m to 120 m).

Second, this study quantified the relative importance and nonlinear relationships with LST between neighborhood LST and both external and internal factors. In this study, the building coverage index in the buffer zone (BCI_buffer) was among the most important variables. BCI_buffer, TCC_buffer, construction year (CY), BCI, TCC, and BAH consistently ranked as the top six influential factors across different weather scenarios. These factors exhibit complex nonlinear relationships with LST; in particular, CY displays a distinct threshold effect in the year 2000.

Third, this study reveals divergent spatial heterogeneity in the relationships between LST and internal versus external neighborhood factors. Internal factors exhibit relatively strong spatial stability, whereas external factors show stronger spatial heterogeneity overall. CY also demonstrates strong spatial heterogeneity under several scenarios. The bandwidths of variables also vary with weather conditions, and the bandwidth of BCI_buffer shrinks sharply during heatwaves. The coefficients of the various factors exhibit instability under various weather conditions.

This study demonstrates that optimizing external spatial planning can mitigate neighborhood heat effects, deepening the theoretical understanding of urban thermal environment differentiation mechanisms. The findings provide scientific evidence for residential planning aimed at combating summer extreme heat and offer practical guidance for old neighborhood renovation and urban renewal initiatives. Particularly in space-constrained inner-city areas, the synergistic regulation of internal and external environments presents a viable strategy for achieving cooling benefits.

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Data availability

The High resolution satellite images were freely accessible via Bigemap Pro (<http://www.bigemap.com/>). The land surface temperature data were retrieved from the Landsat 8/9 Level-2 Science Product (LST band), which is freely accessible via Google Earth Engine (<https://developers.google.com/earth-engine/>). Tree cover data were derived from High Resolution 1m Global Canopy Height Maps (https://gee-community-catalog.org/projects/meta_trees/). Building data were obtained from the Baidu map open platform (<https://lbsyun.baidu.com>). Waterbody and road data are freely accessible from the Open Street Map (<https://www.openstreetmap.org/>). Artificial surface, Vegetation, Barren land data were obtained from the Sentinel-2 Land Cover Explorer(2020)(<https://livingatlas.arcgis.com/landcoverexplorer>). The data of this study is available at <https://github.com/juanfenger/manuscript-2>.

Declaration of Competing Interest

The authors declare no conflict of interest.

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